

Cait Harrigan, MSc.

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I am a PhD candidate at the University of Toronto supervised by Quaid Morris and Kieran Campbell. I am a Doctoral Fellow at the UofT Data Sciences Institute and a graduate researcher at the Vector Institute and Ontario Institute for Cancer Research. I use machine learning to understand cancer genomics by modelling the evolutionary constraints that underlie how mutations occur in DNA. I'm passionate about open science, and promoting great mentorship in the sciences.

EDUCATION

PhD in Computer Science, University of Toronto <i>Supervised by Quaid Morris and Kieran Campbell</i>	01/21 - present
MSc. in Computer Science, University of Toronto <i>Supervised by Quaid Morris</i>	09/19 - 03/21
Honours BSc. in Computational Biology, University of Toronto <i>Awarded with distinction</i>	09/15 - 06/19

RESEARCH EXPERIENCE

Visiting graduate researcher The Francis Crick Institute, London, England <i>Hosted by Nicholas McGranahan</i>	03/24 - 05/24
Visiting graduate researcher Memorial Sloan Kettering Cancer Center, New York, USA <i>Hosted by Quaid Morris</i>	05/21 - 09/21
Undergraduate research assistant SickKids Hospital, Toronto, Canada <i>Supervised by Michael Wilson and Anna Goldenberg</i>	05/17 - 09/17
Intern - Pharmacoeconomics Eviviz, Vancouver, Canada	05/16 - 09/16

ACADEMIC TALKS

Mutational Signatures for DNA Damage and Misrepair , BIRS: Mathematical Methods in Cancer Biology, Evolution and Therapy, <i>Invited talk</i>	05/23
DAMUTA: Dirichlet allocation of mutations as a function of both damage and DNA repair , Cold Spring Harbour Laboratory Meeting: Genome Informatics, <i>Selected Talk</i>	11/21
TrackSigFreq: subclonal reconstructions based on mutation signatures and allele frequencies , Pacific Symposium on Biocomputing, <i>Selected Talk, Poster</i>	01/20

PEER REVIEWED PUBLICATIONS

* Indicates joint first authorship, † co-corresponding authorship

Regional Mutational Signature Activities in Cancer Genomes

Caitlin Timmons, Quaid Morris[†], and [Caitlin F. Harrigan](#)[†]

PLOS Computational Biology, 2022

Activating mTOR mutations are detrimental in nutrient-poor conditions

Agata A. Bielska, [Caitlin F. Harrigan](#), Yeon Ju Kyung, Quaid Morris, Wilhelm Palm, and Craig B. Thompson

Cancer Research, 2022

Reconstructing evolutionary trajectories of mutation signature activities in cancer using TrackSig

Yulia Rubanova, Ruian Shi, [Caitlin F. Harrigan](#), Roujia Li, Jeff Wintersinger, Nil Sahin, Amit Deshwar, and Quaid Morris

Nature Communications, 2020

TrackSigFreq: subclonal reconstructions based on mutation signatures and allele frequencies

[Caitlin F. Harrigan](#), Yulia Rubanova, Quaid Morris, and Alina Selega

Pacific Symposium on Biocomputing, 2020

OTHER PUBLICATIONS

Measuring Scientific Capabilities of Language Models with a Systems Biology Dry Lab

Haonan Duan*, Stephen Zhewen Lu*, [Caitlin F. Harrigan](#), Nishkrit Desai, Jiarui Lu, Michał Koziarski, Leonardo Cotta, Chris J. Maddison

Workshop paper at ICML 2025

Evolutionary characterisation of lung cancer metastasis

Sonya Hessey*, Abigail Bunkum*, Ariana Huebner*, Kristiana Grigoriadis*, Kerstin Haase*, Cristina Naceur-Lombardelli*, Wing Liu, [Caitlin F. Harrigan](#), Daniele Marinelli, Boyue Ding, Corentin Richard, Carlos Martinez-Ruiz, Mark Hill, Olivia Lucas, Oriol Pich, Kerstin Thol, Piotr Pawlik, Takahiro Karasaki, Sophia Ward, Foteini Athanasopoulou, Monica Sivakumar, Selvaraju Veeriah, Antonia Toncheva, Andrew Rowan, Paulina Prymas, Hayley Bridger, Miriam Mitchinson, Elaine Borg, Mary Falzon, Ian Proctor, Ula Mahadeva, Anna Green, Martin Forster, Sarah Benafif, Tanya Ahmad, Siow Lee, Dionysis Papadatos-Pastos, Babu Naidu, Gerald Langman, Matthew Krebs, Pedro Oliveira, Fiona Blackhall, Yvonne Summers, Jamie Weaver, John Le Quesne, Anne Thomas, Cathy Richards, Dean Fennell, Sanjay Jogai, Judith Cave, Patricia Roxburgh, Sioban Fraser, Alan Kirk, Kevin Blyth, Peter Russell, Crispin Hiley, Allan Hackshaw, The TRACERx consortium, TRACERx EVO Consortium, PEACE Consortium, Simone Zaccaria, Nicholas McGranahan[†], Charles Swanton[†], Mariam Jamal-Hanjani[†]

Under consideration at Nature

Damage and Misrepair Signatures: Compact Representations of Pan-cancer Mutational Processes

[Caitlin F. Harrigan](#), Kieran R. Campbell, Quaid Morris[†], Tyler Funnell[†]

Under consideration at Nature Communications

Automated registration of spatial expression data scales multimodal integration to large cohorts

[Caitlin F. Harrigan](#), Ching Yeung Lam, Danian Chen, Christian Lai, Rod Bremner, Hartland W. Jackson, Kieran R. Campbell

Under consideration at Cell Systems

Whole slide Imaging Mass Cytometry allows the rapid profiling of the immune landscape of histopathologically aggressive prostate tumors

Jennifer L. Gorman, Lydia Y. Liu, Jordan P. Hartig, Nikesh Parsotam, Amanda Khoo, Vladimir Ignatchenko, Sarah Asbury, Somi Afiuni, Ricardo Gonzalez, Michael J. Geuenich, Caitlin F. Harrigan, Yuju Lee, Jianan Chen, Liang Lim, Qanber Raza, Peggi M. Angel, Kieran Campbell, Stanley K. Liu, Michelle R. Downes, Richard R. Drake, Thomas Kislinger, David King, Hartland W. Jackson

AACR annual meeting proceedings, 2024

Bacterial ADP-heptose initiates a revival stem cell program in the intestinal epithelium

Shawn Goyal, Cynthia X. Guo, Adrienne Ranger, Derek K. Tsang, Ojas Singh, Caitlin F. Harrigan, Olga Zaslaver, Hannes L. Rost, Herbert Y. Gaisano, Scott A. Yuzwa, Nan Gao, Jeffrey L. Wrana, Dana J. Philpott, Scott D. Gray-Owen and Stephen E. Girardin

Cell Stem Cell, 2025

Considerations for Visualizing Uncertainty in Clinical Machine Learning Models

Caitlin F. Harrigan*, Gabriella Morgenshtern*, Anna Goldenberg, and Fanny Chevalier

Workshop paper at CHI 2021

POSTERS

Mapping the Evolution of Mutational Signatures on Tumour Phylogenies, ISMB 2024 07/24

Dirichlet Allocation of Mutations Captures the Action of DNA Damage and Misrepair Processes, ISMB 2022 07/22

TrackSigFreq: subclonal reconstructions based on mutation signatures and allele frequencies, MLCB 2021 12/19

ACADEMIC AWARDS

NSERC Postgraduate Scholarship - Doctoral 09/22 - present
Natural Sciences and Engineering Research Council of Canada

Doctoral Student Fellowship Award 09/22 - present
Data Science Institute, University of Toronto

Graduate Research Award 03/24 - 05/24
Mitacs, in partnership with UKRI

Queen Elizabeth II Graduate Scholarship in Science & Technology 09/22
Province of Ontario
Respectfully declined

Ontario Graduate Scholarship 09/21 - 09/22
Province of Ontario

ACM SIGHPC Computational & Data Science Fellowship 07/20 - 07/22
Special Interest Group on High Performance Computing of the Association for Computing Machinery

JXTX foundation Genome Informatics Scholarship 08/21
James P. Taylor Foundation for Open Science

General Motors Women in Science and Mathematics Award 09/20
University of Toronto

NIH Conference Travel Fellowship	11/19
International Society for Computational Biology	
The Audrey Taylor Award for co-curricular contribution to the New College community	07/17
University of Toronto	

GRANTS

Emergent Data Sciences Program Grant	09/24
Data Science Institute, University of Toronto	
<i>Awarded for CrossTALK drug discovery bootcamp</i>	
Student initiative fund	02/18
University of Toronto	
<i>Awarded for Bioinformatics and Computational Biology Hackathon</i>	
Event grant	02/18
New College Student council	
<i>Awarded for Bioinformatics and Computational Biology Hackathon</i>	

ACADEMIC SERVICE

Jouranal peer review: Genome Biology (2023), Genome Medicine (2021), Nature Cancer (2024), PLOS Computational Biology (2023, 2023), iScience (2021)

Conference referee: ICML 2025 workshop CODEML (2025)

Conference program committee: Machine Learning in Computational Biology (2019), Pacific Symposium on Biocomputing (2020)

STUDENT SUPERVISION

Kiki Zhang , Combined BS/MSE Student; Biomedical Engineering. Research internship via Computational Biology Student Program at MSKCC, <i>Topic: Mutational signatures in the context of branching evolution</i>	06/23 - 08/23
Fedir Zhydok , BS Student; Computer Science. Artificial intelligence in medicine, global classroom program at University of Toronto, <i>Topic: Identifying metastatic tumours from mutational signatures</i>	09/22 - 12/22
Caitlin Timmons , BA Student; Statistical and Datasciences, Biology. Research internship via Computational Biology Student Program at MSKCC. Went on to a Research Technician position at Dana-Farber Cancer Institute., <i>Topic: Modelling spatial distribution of mutational signatures in cancer genomes.</i>	05/21 - 08/22
Haritha Lakshmanan , Highschool Student. Independent study at MSKCC. Went on to a combined BA/MD at Brooklyn College., <i>Topic: Automatic discovery of mutations predictive of survival in breast cancer patients</i>	05/20 - 11/20

GUEST LECTURES

Feature engineering & data loaders , CrossTALK drug discovery bootcamp	06/25
Exploring & Explaining Data in the Wild , AI & Data Science Post-Graduate Program, Loyalist College, <i>Invited by Prof. Peter Papadakos as an anual guest lecturer in 2023 & 2024</i>	03/23
Data Collection & Analysis , PRISM research & mentorship program, University of Toronto, <i>Invited by Prof. Sadia Sharmin</i>	02/22
Environmental & Life Sciences Workshop Series , STEMHub Foundation	01/20
R for bioinformatics workshop , Global Society for Genetics and Genome Biology	01/20

OTHER TALKS

Machine Learning for Cancer Genomics , Artificial Intelligence in Healthcare Society Case Competition, York University, <i>Invited keynote</i>	03/24
Finding the ‘I’ in science , ACM Canadian Celebration of Women in Computing, <i>Selected Workshop</i>	10/22
Undergraduate research opportunities: how to find them and make them work for you , UofT Bioinformatics and Computational Biology Student Union, <i>Invited by the BCBSU</i>	02/20
How to hack your degree , Computer Science Student Union, University of Toronto, <i>Invited by the CSSU</i>	05/19

PROGAM & EVENT ORGANIZATION

Program Organizer UofT Graduate Application Assistance Program	09/21 - 01/25
Project Manager STEMHub Foundation, Toronto, Canada	10/20 - 05/21
Volunteer Coordinator Computer science grad visit day, University of Toronto	03/20
Founder and treasurer Bioinformatics and Computational Biology Student Union, University of Toronto	05/18 - 05/19
Program information session UofT Bioinformatics and Computational Biology Student Union	06/17 - 06/19
Planning executive Bioinformatics and Computational Biology Hackathon 2018	03/18

PROFESSIONAL TRAINING

The Carpentries certified instructor	10/23
Oxford ML summer school (Health track)	07/23

TEACHING ASSISTANT POSITIONS

Unless otherwise noted, school is University of Toronto

CSC2541: Topics in Machine Learning: AI for Drug Discovery , Head prep TA	04/25 - present
CrossTALK drug discovery bootcamp , Head TA	01/25 - present
JSC370: Data Science II	01/23 - 05/23
JSC270: Data Science I	01/23 - 05/23
STA313: Data Visualization	09/22 - 12/22
PRISM: Preparation for Research through Immersion, Skills, and Mentorship	01/22 - 05/22
	01/22 - 05/22
CSC197: Privacy in the Age of Big Data Collection	09/21 - 12/21
STA4273: Minimizing Expectations	01/21 - 05/21
CSC197: Privacy in the Age of Big Data Collection	09/20 - 12/20
JSC270: Data Science I	01/20 - 05/20
CSC373: Algorithm Design, Analysis & Complexity	09/19 - 12/19

MENTORSHIP ACTIVITIES

As part of my ongoing commitment to supporting students at all levels and background in engaging with computational biology, I make an effort to be available to provide guidance and resources to students, with a particular focus on creating an inclusive environment that fosters diverse perspectives and experiences. In addition to being a mentor through the organized programs listed here, I set aside ~2h/month for by-request 30 minute meetings.

Graduate Application Assistance Program , University of Toronto	09/21 - 01/25
Computer Science Alumni Mentorship Program , University of Toronto	01/23 - 05/23
Statistical Science Alumni Mentorship Program , University of Toronto	10/21 - 05/22
ProjectX machine learning research competition , UofT AI club	09/20 - 02/21
Her Code Camp , Toronto, Canada	08/20 - 08/21
Department of Statistics Mentorship Program , University of Toronto	10/18 - 05/19
SPROUT Peer Mentorship Program , University of Toronto	09/16 - 05/19
New College E-Mentorship Program , University of Toronto	09/16 - 05/19

HACKATHONS

Silver medal , Toronto Bioinformatics Hackathon 2024	09/24
Silver medal , American Statistical Association DataFest Competition	05/18

COMMUNITY ACTIVITIES

Webmaster , Project Galiwatch	03/20 - present
Graduate student ambassador , Computer science grad visit day, University of Toronto	03/25
Judge , Lunenfeld-Tanenbaum Research Institute Summer Poster Session	07/24
Webmaster , IMERSS Xetthecum ecocultural digital mapping project	10/23 - 12/24
Volunteer , Meals on wheels	06/23 - 09/23
Volunteer , Billion oysters project	05/21 - 09/21
Communications and Marketing Executive , UofT Women in Computer Science	09/16 - 05/17

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